

Fatih Jawwad Al-Mumtaz

+62-8514-3396-221 | wedjaw22@gmail.com | linkedin.com/in/fatih-al-mumtaz | github.com/jaweed3 | Madura, Indonesia

SUMMARY

ML engineer with merged PRs in the Burn deep learning framework and a track record of shipping production systems under tight constraints: recommendation APIs at <1ms latency, anomaly detection on ESP32-S3 at 256KB SRAM, and MLOps pipelines cutting drone SAR deployment to 15 minutes. Erasmus+ scholar. Writes Rust and Python that ships.

EDUCATION

University of Darussalam Gontor (UNIDA) <i>Bachelor of Computer Engineering — GPA: 3.7/4.0</i>	Ponorogo, Indonesia <i>Aug 2024 – Present</i>
Erasmus+ Exchange (KA171) Universidad de Sevilla <i>Design Testing & QA, Signal Processing, Numerical Modeling & Simulation</i>	Seville, Spain <i>Sep 2025 – Feb 2026</i>
Gontor Islamic Boarding School <i>Graduated with Distinction (Mumtaz)</i>	Ponorogo, Indonesia <i>2018 – 2024</i>

EXPERIENCE

Google Student Ambassador <i>Campus Ambassador</i> - Led 15+ ML/data science workshops, reaching 100+ students. - Connected 6 campus groups with Google tools and industry resources.	UNIDA Gontor, Indonesia <i>2024 – Present</i>
Co-Founder, Forum DA'I (Darussalam Artificial Intelligence) <i>Founder & Lead Organizer</i> Founded campus's first AI club; ran 20+ weekly deep learning sessions.	UNIDA Gontor, Indonesia <i>2024 – Present</i>
IT Support Staff <i>Systems & Infrastructure</i> Maintained 50+ workstations and library system serving 5,000+ students.	UNIDA Gontor Library, Indonesia <i>2024 – Present</i>

PROJECTS

burn-recsys <i>Rust, Burn, Axum, Polars, OpenTelemetry, HNSW</i> - Built 3 recommendation architectures (GMF, NeuMF, DeepFM) with <1ms inference latency. - Designed scalable data pipeline and production Axum API serving 1,000+ req/s at <1ms p50. - Containerized with CI/CD; achieved 0.604 HR@10 and 0.414 NDCG@10 on Myket.	2026
Tracel AI — Burn (Open Source) <i>Rust, Burn</i> - Contributed ROUGE-L metric (283 lines, 2 files) to Burn's training crate. - Fixed SgdConfig initialization bug, merged upstream.	2026
vibeSentinel <i>Rust, ESP32-S3, Autoencoder, Embedded</i> - Deployed autoencoder for vibration anomaly detection on ESP32-S3 under 256KB SRAM. - Built Rust toolchain compressing 2MB models to fit sub-MB SRAM.	2026
retakId <i>TensorFlow.js, Edge ML, Delta OTA, Telegram Bot</i> - Shipped TensorFlow.js edge model detecting landslide cracks from 4 inputs. - Delta OTA reduces model updates by 80%; Telegram bot serves 100+ offline users. - Won Best Thematic Award among 40+ teams at IYREF ITB 2026.	2026
RescueVision Edge <i>YOLOv8n, ONNX INT8, TFLite, FastAPI, MLflow, DVC</i> Built MLOps pipeline cutting SAR drone deployment to 15 minutes.	2026
BitJETS-M4 (bits) <i>PyTorch, BitNet b1.58, MPS, HiFi-GAN</i> - Fine-tuned 1.58-bit quantized TTS model compressing 350MB to 12MB on Apple M4. - Combined quantized acoustic model with FP32 vocoder under 8GB VRAM.	2025

AWARDS & ACHIEVEMENTS

3rd Place — FTI Hackathon, Mobile Apps Dev Universitas Kristen Satya Wacana	2025
Erasmus+ KA171 Scholarship European Union	2025 – 2026
Google Student Ambassador UNIDA Gontor	2024 – Present

PUBLICATIONS

Identification of Prayer Movements Using CNN and Prewitt & Morphology Image Processing	2024
Published CNN prayer posture classifier achieving 94% accuracy.	

TECHNICAL SKILLS

Languages: Rust, Python, C++, Kotlin, Bash, TypeScript
ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, TFLite, Burn
Edge & Embedded: ESP32-S3, ESP-IDF, ARM Cortex-M, ONNX INT8, TFLite INT8, OpenVINO
MLOps & Backend: MLflow, DVC, Docker, Kubernetes, FastAPI, Axum, Prometheus, Grafana, OpenTelemetry
Languages Spoken: Indonesian (Native), Madurese (Native), English (Fluent), Arabic (Fluent)